## MASTER WARNING LIGHT

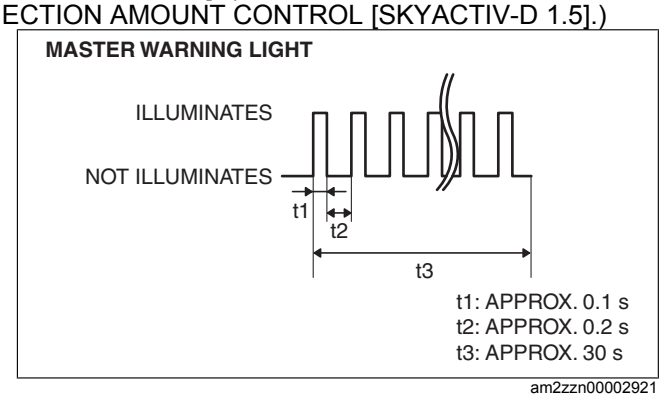
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### Purpose

- The master warning light warns the driver that any of the following malfunctions is occurring.
  - — Brake switch malfunction
  - — Oil pressure switch malfunction (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))
  - — Engine oil solenoid valve malfunction (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
  - — Engine oil pressure sensor malfunction (SKYACTIV-D 1.5)
  - — Engine oil level sensor malfunction (SKYACTIV-D 1.5)
  - — Power supply system malfunction
  - — Current sensor malfunction (With i-stop or i-ELOOP)
  - — Global central configuration error
  - — Timing chain malfunction (SKYACTIV-D 1.5)
  - — Blow-by heater malfunction (SKYACTIV-D 1.5)
  - — i-ELOOP system malfunction (With i-ELOOP)

### Function

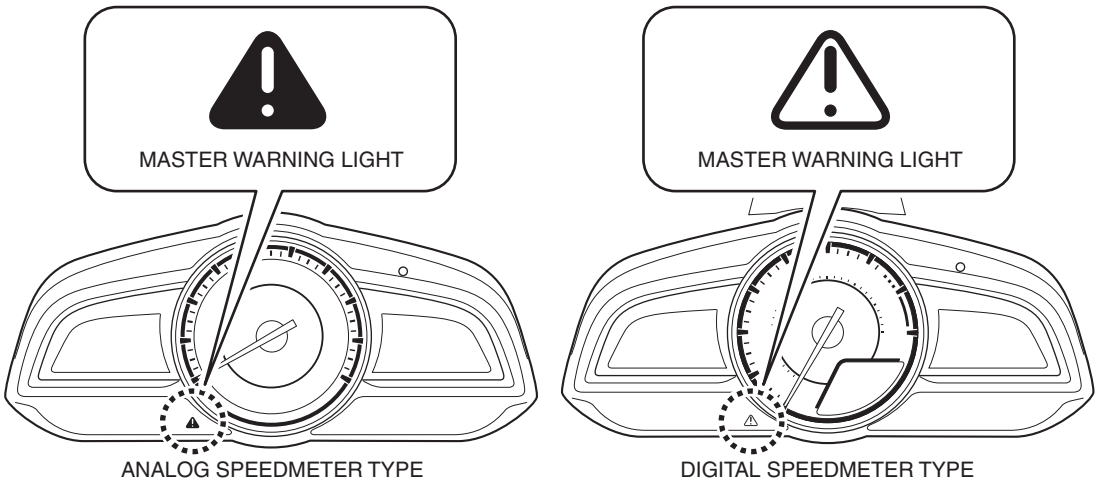
- The instrument cluster turns the master warning light on based on the following CAN signals sent from the PCM.
  - — Brake switch No.1 error signal
  - — Brake switch No.2 error signal
  - — Oil pressure switch error signal (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))
  - — Engine oil solenoid valve error signal (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
  - — Engine oil pressure sensor error signal (SKYACTIV-D 1.5)
  - — Engine oil level sensor error signal (SKYACTIV-D 1.5)
  - — Power supply system error signal
  - — Current sensor error signal (With i-stop or i-ELOOP)
  - — Global central configuration error signal
  - — Timing chain error signal (SKYACTIV-D 1.5)
  - — Blow-by heater error signal (SKYACTIV-D 1.5)
  - — i-ELOOP system error signal (With i-ELOOP)
- If the cancel procedure for preventing unnecessary operation of the brake override system is implemented, the PCM sends a master warning light flash request signal to the instrument cluster via CAN communication. The instrument cluster flashes the master warning light based on the master warning light flash request signal from the PCM. For the cancel procedure to prevent unnecessary operation of the brake override system, refer to [DRIVE-BY-WIRE CONTROL] [FUEL INJECTION AMOUNT CONTROL]. (See DRIVE-BY-WIRE CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See FUEL INJECTION AMOUNT CONTROL [SKYACTIV-D 1.5].)
- The master warning light flashes in the pattern shown in the figure.




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### Construction

- Turning on/flashing/turning off of the master warning light is controlled by the instrument cluster microcomputer based on each error signal or master warning light flash request signal sent from the PCM.
- The master warning light is built into the instrument cluster.

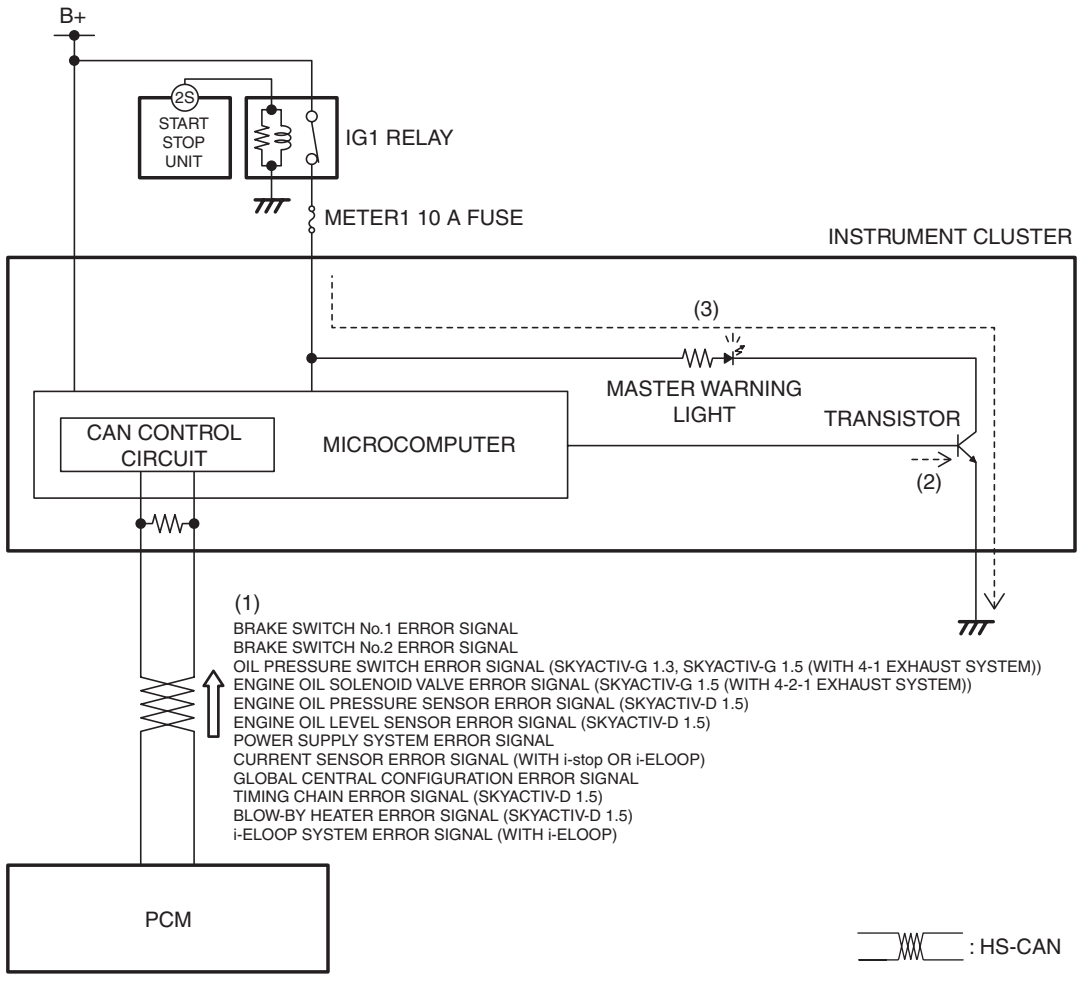



**INSTRUMENT CLUSTER**

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**Operation**

1. When the ignition is switched ON (engine off or on), the instrument cluster receives (1) each error signal or a master warning light flash request signal from the PCM via the CAN communication.
2. The instrument cluster turns the transistor on (2) intermittently for flashing and continuously for illumination based on each signal.
3. The master warning light flashes (3) when the transistor is turned on intermittently, and it turns on (3) when the transistor is turned on continuously.




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**Fail-safe**

- Not applicable